

alphabetic ciphers. This was later redeveloped and published by Friedrich Kasiski, a Prussian cryptologist.

Kerckhoff's Principle

During that time, cryptography was mainly a culmination of basic rules, developed and reiterated over time. Frenchman Auguste Kerckhoffs' essays on cryptography laid the cornerstone for military use of the science. These articles scrutinized the state-of-the-art military cryptography of the time and appealed for massive improvements in the French practice. Apart from practical advice and basic rules, it also had six principles of cipher design for the practical age.

These principles emphasized on the simplicity of the key (which could be memorized) and the belief that the cipher should be unbreakable and transmittable by telegraph. Furthermore, it also stated that the documents



Skytale

should be handled by a single person and the system should be easy, without any requirement of mental strain. However, it was his second principle that became quite popular and came to be known as Kerckhoffs' principle. It stated that, "The design of a system should not require secrecy and compromise of the system should not inconvenience the correspondents." This gave an impetus to cryptography and spurred further milestones.

The misuse of cryptography led to the execution of Mata Hari in the early 20th century. Similarly, during World War I, the German naval codes were broken by the Admiralty's Room 40 (British Naval Intelligence), which perhaps played an important role in detecting German activity in the North Sea, giving Britain an advantage in the battles of Dogger Bank and Jutland.

Black Chamber

Cryptanalysis units sprouted across Europe and were known as the Black Chamber. It started off in 1628 when Frenchman Antoine Rossignol helped the French army by deciphering a captured message and subsequently, defeated the Huguenots. The victory ensured that Rossignol was called upon to decode ciphers for the Government. His method of deciphering the